# Zhongxiao (John) Liu

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## Education

**Duke University** Expected May 2028
Master of Science in Mechanical Engineering & Materials Science Durham, NC

**Donghua University** Sep 2022 - Jun 2026
Bachelor of Engineering in Industrial Design Shanghai, China

## Technical Skills

**Mechanical Design:** SolidWorks, FEA Stress Analysis, Fastener Selection, Mechanical BOMs

**Fabrication & Assembly:** Sheet Metal Flat Patterns, Spot Welding, On-Site Mechanical Rework

**Robotics & Systems:** ROS 2, Jetson Linux, Kernel Compilation, LiDAR/Camera Integration

**Wiring & Prototyping:** Custom Harness Design, Cable Routing, Braided Sleeving, SMD Microsoldering

## Projects

**Heavy-Duty Hydraulic Safety Valve Test Platform** Sep 2025 - May 2026

Undergraduate Capstone | Donghua University

- Designed a custom safety-valve test platform; delivered a mechanical BOM and fabrication-ready STEP files to vendors.
- Compared three clamping concepts using FEA stress analysis and selected a three-jaw design based on gripping requirements and material cost.
- Resolved rail/frame interference during on-site assembly through mechanical rework, avoiding a redesign of the fabricated platform.

**Autonomous Mobile Robot - Hardware & System Integration** Jan 2026 - Aug 2026

Independent Project

- Integrated an NVIDIA Jetson Orin Nano, LiDAR, and Orbbec camera with an STM32 controller on a Mecanum-wheel robot.
- Configured Jetson Linux, compiled the kernel, and built ROS 2 from source to support robot hardware integration.

**SafeStep - Wearable Load Monitoring System** Sep 2026 - Present

Hardware Lead | Duke University Course Project | In Progress

- Reviewed resistive, capacitive, and strain-gauge sensor options and defined requirements for an eight-point wearable load-monitoring prototype.
- Developing a ZNX-01-based data acquisition device for wearable load monitoring; responsible for sensor procurement, BOM preparation, calibration, and sensor data streaming.

**Robotic U-Bend Assembly - Gripper Optimization** Sep 2026 - Present

Industry-Partnered Course Project | Duke University x Trane Technologies | In Progress

- Responsible for gripper optimization, with design objectives covering level grasping, clearance at corner locations, and copper-tube protection.

## Additional Experience

**Independent Hardware Repair & Maker** 2019 - 2026

Freelance | Shanghai, China

- Performed SMD-level diagnostics and microscope-assisted soldering on logic boards and wearables; restored industrial inkjet printheads through cleaning and recalibration.
- Built an audio system using salvaged drivers, Class-D amplification, and custom connectors.